



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

SECP3133-01

HIGH PERFORMANCE DATA PROCESSING

Assignment 1 Logbook

LECTURER: DR. MOHD SHAHIZAN BIN OTHMAN

NO.	NAME	MATRIC NUMBER
1.	IMAN ABADI BIN MOHD NIZWAN	A23CS0084

Date	Search Keywords	AI Prompts Used	Task Description
6 April 2026	Apache Spark Streaming on WebOfScience (WoS)	None	Searched WoS for papers on Apache Spark Streaming. Found 5 relevant articles. Selected 2 to read.
10 April 2026	Apache Flink on WoS	None	Searched WoS for papers on Apache Spark Streaming. Found 5 relevant articles. Selected 3 to read.
12 April 2026	None	Find peer-reviewed research papers that compare stream processing frameworks such as Apache Spark Streaming and Apache Flink. Focus on studies that evaluate performance metrics such as latency, throughput, scalability, fault tolerance, and resource utilization in real-time big data processing systems. Prefer papers published after 2021 in journals or conferences related to big data, distributed systems, or data engineering using Scopus AI	Used Scopus AI to find papers that include comparison metrics on both Apache Spark Streaming and Apache Flink. Selected Found and 11 related articles.
14 April 2026	None	None	Created a draft comparison table comparing both frameworks in terms of streaming model, latency, throughput, fault tolerance and scalability.

17 April 2026	None	Give me a draft on the structure of the paper based on the assignment briefing file. Provide points I should include based on the topic given suing ChatGPT	Write introduction overview on Apache Spark Streaming.
18 April 2026	None	Explain to me on the Apache Spark Streaming architecture using Claude	Drew an architecture diagram using Lucid chart and explain it. Write on real-word case study on using Apache Spark Streaming for health status prediction.
19 April 2026	None	None	Create a comparison table and compare on streaming model, latency and throughput of both frameworks.
22 April 2026		Based on this paper, explain why micro-batch models like Apache Spark Streaming lacks operator-level fault tolerance while native-streaming model does not. Relate with the types of operators (stateful and stateless) and answer which model is more fault tolerant and why using Claude	Explain on comparison for fault tolerance and scalability.
25 April 2026	None	None	Write on the abstract, conclusion and keywords. Format and finalize the paper
26 April 2026	None	None	Submitted the paper on Turnitin. Made changes on flagged sentences
27 April 2026	None	None	Resubmitted the paper and finalize the final submission

